Raffles Institution
Physics Department
Revision Package 1

Circuits

16

Important note: ****denotes questions that may require knowledge from later chapters in order to solve them.***

I. MCQs

1. **[AJC/H2/Prelims 2010/P1/26]**

A moving coil ammeter whose resistance is 0.50 Ω registers a current of 3.0 A when connected in series with an unknown resistor *R* and a battery of e.m.f. 12 V with negligible internal resistance.

What is the percentage error in the reading of the ammeter?

A 9.0% B 13% C 20% D 88%

2. **[CJC/H2/Prelims 2010/P1/26]**

Two wires P and Q, each of the same length and the same material, are connected in parallel to a battery. The diameter of P is half that of Q.

What fraction of the total current passes through P?

A 0.20 B 0.25 C 0.33 D 0.50

3. **[Dunman High/H2/Prelims 2010/P1/24]**

Two cylindrical resistors, one made of copper and the other of aluminum, have the same volume. The cross-section area and the resistivity of aluminum are both three times that of copper.

What is the magnitude of the ratio $\dfrac{\text{resistance of aluminium resistor}}{\text{resistance of copper resistor}}$?

A 9 B 3 C $\dfrac{1}{3}$ D $\dfrac{1}{9}$

4. **[MI/H2/Prelims 2010/P1/25]**

What is the definition of resistance?

A Resistance is the potential difference per unit current.

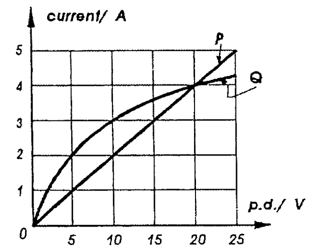
B Resistance is the gradient of the graph of potential difference against current.

C Resistance is the voltage required for a current of 1 A.

D Resistance is defined by the equation $R = \dfrac{\rho l}{A}$, where ρ is the resistivity of the material, l is the length of the wire, and A is its cross-sectional area.

5. **[Dunman High/H2/Prelims 2010/P1/25]**

The figure shows the graph of current against potential difference for two electrical devices P and Q.

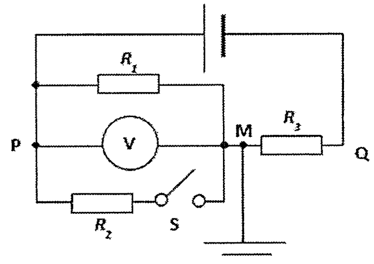


P and Q are joined in series. If the potential difference across P is 10 V, which of the following is correct?

	p.d. across Q / V	p.d. across P and Q / V
A	5.0	10
B	5.0	15
C	15	15
D	15	25

6. **[HCI/H2/Prelims 2010/P1/26]**

A thermistor R_1 is connected to a battery of constant e.m.f. with negligible internal resistance as shown.



Which of the following actions will cause an increase in the potential difference *V* measured by the voltmeter? Assume that the voltmeter has infinite resistance.

A Increase the temperature of the thermistor with S open

B Remove the earth connection at M with S open

C Close switch S

D Decrease resistance R_3 with S open

7. **[IJC/H2/Prelims 2010/P1/26]**

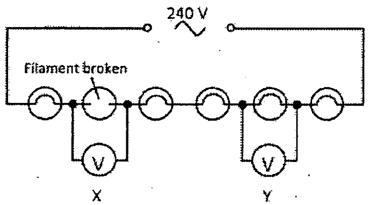
The resistivity of aluminum is 2.0 times that of silver. An aluminium wire of length *L* and diameter *d* has a resistance *R*.

What is the diameter of the silver wire, also of length *L* and resistance *R*?

A 0.05 *d* B 0.71 *d* C 1.4 *d* D 2.0 *d*

8. **[MI/H2/Prelims 2010/P1/26]**

A main circuit has six identical bulb connected in series. One of the bulbs has a broken filament. Voltmeters **X** and **Y** of infinite resistance are placed in the circuit as shown.



What are the voltmeter readings?

	X reading	Y reading
A	0 V	0 V
B	0 V	240 V
C	40 V	40 V
D	240 V	0 V

9. **[MJC/H2/Prelims 2010/P1/26]**

The circuit shown in Fig. 9.1 may be used to determine the internal resistance of a battery. An oscilloscope is connected across the battery as shown. Fig. 9.2 represents the screen of the oscilloscope.

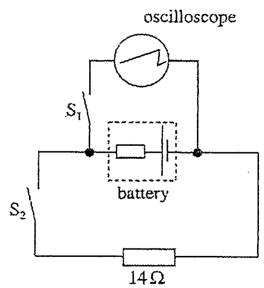


Fig. 9.1

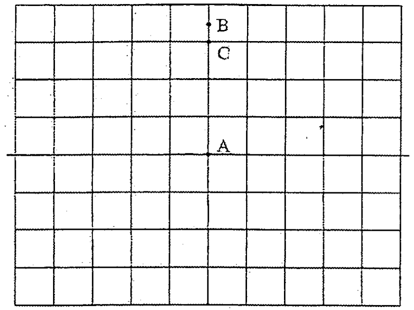


Fig. 9.2

The time base of the oscilloscope is switched off throughout the experiment.

Initially the switches **S_1** and **S_2** are both open. Under these conditions, the spot on the oscilloscope screen is at **A**.

Switch **S_1** is now closed, with **S_2** remaining open. The spot moves to **B**.

Switch **S_1** is kept closed and **S_2** is also closed. The spot moves to **C**.

The vertical sensitivity of the oscilloscope is 0.50 V per division.

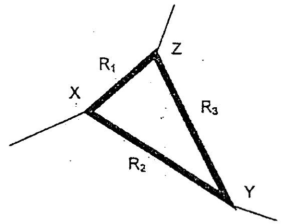
What is the internal resistance of the battery?

A 0.24 Ω **B** 2.3 Ω **C** 14.0 Ω **D** 16.4 Ω

10. **[NJC/H2/Prelims 2010/P1/27]**

The figure below shows a network of three resistance wires marked R_1, R_2 and R_3. All three wires have the same cross-sectional area A. Wire R_1 has resistivity ρ and length l. Wire R_2 has resistivity 2ρ and length $2l$, while wire R_3 has resistivity 0.5ρ and length $2l$.

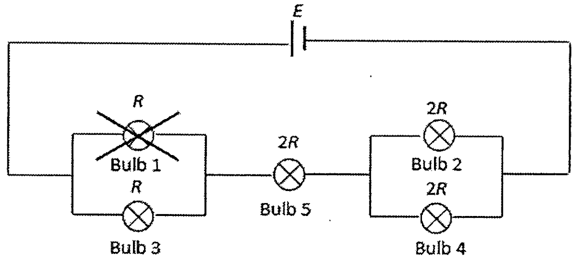
What is the resistance between X and Y?



A $0.75\frac{\rho l}{A}$ **B** $0.83\frac{\rho l}{A}$ **C** $1.3\frac{\rho l}{A}$ **D** $4.0\frac{\rho l}{A}$

11. **[NJC/H2/Prelims 2010/P1/28]**

Five light bulbs with difference resistances as indicated in terms of R in the diagram below are connected to a constant voltage d.c. supply, E.



If bulb 1 blows as shown, what happens to the brightness of the remaining bulbs?

	Bulb 2	Bulb 3	Bulb 4	Bulb 5
A	Decreases	Increases	Decreases	Decreases
B	Increases	Decreases	Increases	Decreases
C	Decreases	Decreases	Decreases	Decreases
D	Decreases	Increases	Decreases	Increases

12. **[NYJC/H2/Prelims 2010/P1/25]**

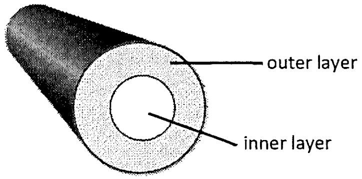
A strain gauge consists of a length of wire with uniform cross-sectional area. Its resistance is 2.000 kΩ. It is attached to a gas container. When the container expands, the strain gauge changes its dimensions. Its length increases by 0.400 % and diameter reduces by 1.00 %.

What is the new resistance of the strain gauge?

A 1.006 kΩ **B** 1.968 kΩ **C** 2.049 kΩ **D** 2.567 kΩ

13. [TJC/H2/Prelims 2019/P1/20]

A cylindrical electrical wire which is 10 m in length consists of an inner cylindrical core made of copper and an outer layer made of aluminium as shown in the figure. The diameter of the inner core is 0.0050 m and the diameter of the whole rod is 0.010 m.



Given that the resistivity of copper is 1.7×10^{-8} Ω m and the resistivity of aluminium is 2.7×10^{-8} Ω m, what is the overall resistance of the electrical wire?

A 0.0030 Ω **B** 0.0060 Ω **C** 0.0090 Ω **D** 0.013 Ω

14. [RI/H2/Prelims 2010/P1/26]

A car battery of e.m.f. 12 V and internal resistance 0.020 Ω is connected to a load of 4.0 Ω.

If the potential difference across the load is 10 V, what is the power lost in the connecting wires?

A 0.13 W **B** 1.0 W **C** 4.9 W **D** 5.0 W

15. [RVHS/H2/Prelims 2010/P1/24]

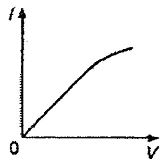
Two cubes, X and Y are cut from the same block of metal. The linear dimension of X is 2 times that of Y.

What is the ratio $\frac{R_X}{R_Y}$ of the resistances between the opposite faces of X and of Y?

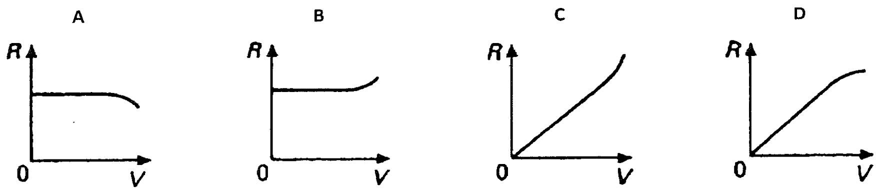
A $\frac{1}{2}$ **B** 1 **C** 2 **D** 4

16. [VJC/H2/Prelims 2010/P1/25]

The current I flowing through a component varies with the potential difference V across it as shown.

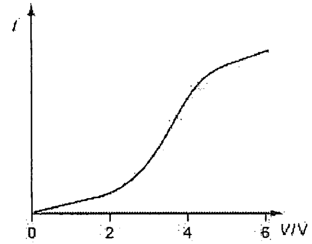


Which graph best represents how the resistance R varies with V?

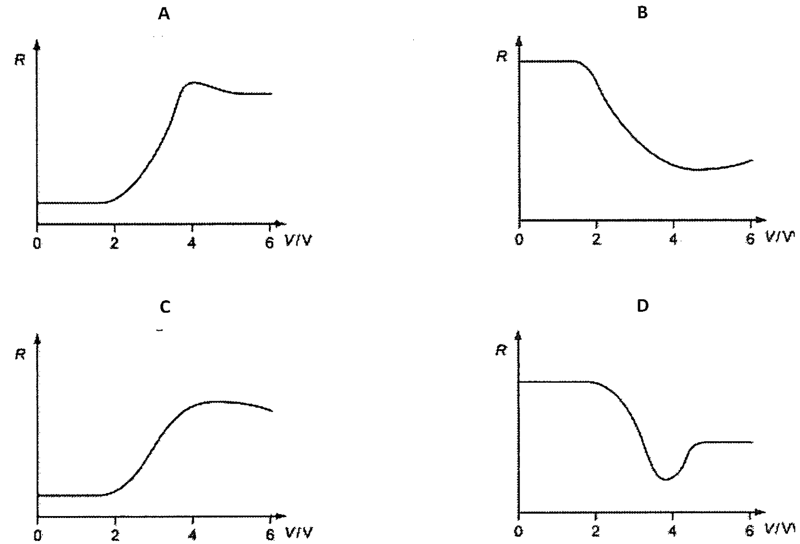


17. [RVHS/H2/Prelims 2010/P1/25]

The electrical characteristic of a component is shown below.

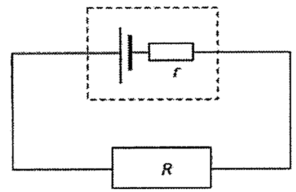


Which graph below shows the way the resistance of the component varies with applied voltage?



18. [HCI/H2/Prelims 2010/P1/27]

A battery with internal resistance r is connected to a resistor R as shown in the figure below. A constant current passes through R. When a charge of 30.0 C passes through the circuit, the heat dissipated in r is 45.2 J and the heat dissipated in R is 88.3 J.

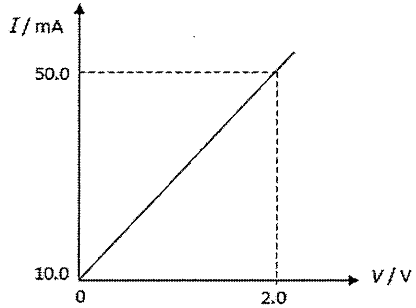


What is the e.m.f. of the battery?

A 0.148 V **B** 1.51 V **C** 2.94 V **D** 4.45 V

19. [TMJC/H2/Prelims 2019/P1/19]

The following graph shows the *I-V* characteristic of a certain conductor.



Which of the following statements about the conductor is correct?

A It obeys Ohm's law and when $V = 2.0$ V, resistance = 40.0 Ω.

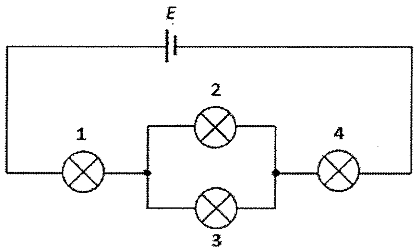
B It obeys Ohm's law and when $V = 2.0$ V, resistance = 50.0 Ω.

C It does not obey Ohm's law and when $V = 2.0$ V, resistance = 40.0 Ω.

D It does not obey Ohm's law and when $V = 2.0$ V, resistance = 50.0 Ω.

20. [YJC/H2/Prelims 2010/P1/26]

Four identical bulbs labeled **1** to **4** are connected with a cell *E* in the circuit shown below.



Which of the followings will increase the brightness of bulb **1**?

A Replace the cell *E* with another cell of lower e.m.f.

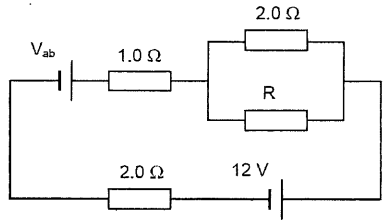
B Remove bulb **2**

C Connect another bulb in parallel with bulb **2**

D Connect another bulb in series with bulb **4**

21. [AJC/H2/Prelims 2010/P1/27]

Current through the resistor R of 1.0 Ω is 2.0 A. What is the terminal voltage V_{ab} of the "unknown" battery? (You may ignore all internal resistance for this question.)

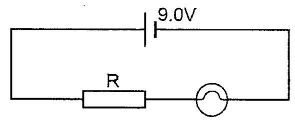


A 1.0 V **B** 2.0 V **C** 10 V **D** 13 V

22. [AJC/H2/Prelims 2010/P1/28]

A flashlight bulb rated at 2.5 W and 3.0 V is operated by a 9.0 V battery. To light the bulb at its rated voltage and power a resistor is connected in series as shown in figure below.

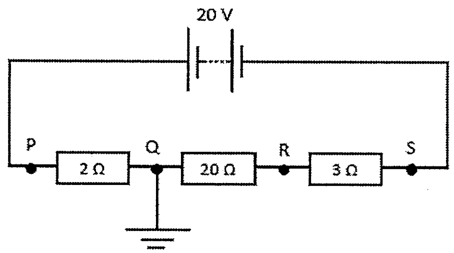
What value should the resistor have? Internal resistance of the battery can be ignored.



A 3.6 Ω **B** 6.3 Ω **C** 7.2 Ω **D** 32 Ω

23. [CJC/H2/Prelims 2010/P1/27]

The diagram shows three resistors of resistances 2 Ω, 20 Ω and 3 Ω connected in series. A potential difference of 20 V is maintained across them. Point Q is earthed.

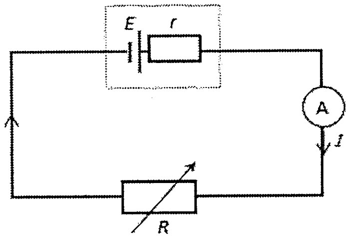


Which of the following gives the potentials at points P, Q, R and S?

	P	Q	R	S
A	20 V	18.4 V	2.4 V	0 V
B	1.6 V	0 V	16 V	18.4 V
C	20 V	16 V	–6 V	–20 V
D	1.6 V	0 V	–16 V	–18.4 V

24. [CJC/H2/Prelims 2010/P1/28]

A battery of e.m.f. E and internal resistance r delivers a current I through a variable resistance R.



R is set at two different values and the corresponding currents I are measured using an ammeter of negligible resistance.

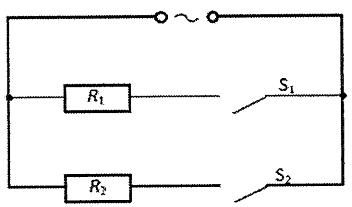
R/Ω	I/A
1.0	3.0
2.0	2.0

What is the value of e.m.f E?

A 3.0 V **B** 3.5 V **C** 4.0 V **D** 6.0 V

25. [CJC/H2/Prelims 2010/P1/29]

An electric heater can be represented as two resistors of resistances R_1 and R_2 and two switches S_1 and S_2. The resistance R_2 is greater than that of R_1.



Which switches must be closed so that the heater produces the maximum possible power and the minimum non-zero power?

	maximum possible power	minimum non-zero power
A	S_1 and S_2	S_2
B	S_1 and S_2	S_1
C	S_1	S_2
D	S_2	S_1

26. [Dunman High/H2/Prelims 2010/P1/26]

Two resistors, R_1 and R_2 are connected in parallel. R_1 has a fixed value and the value of R_2 is variable but always greater than R_1. The combined resistance is

A less than R_1 and increases as R_2 decreases.

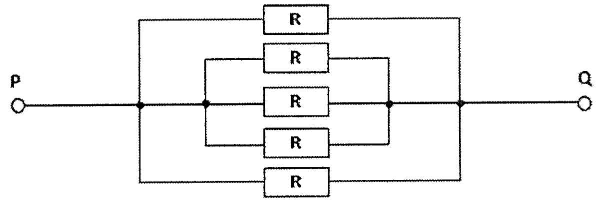
B less than R_1 and decreases as R_2 decreases.

C greater than R_1 and increases as R_2 increases.

D greater than R_1 and decreases as R_2 increases.

27. [YJC/H2/Prelims 2010/P1/24]

Five identical resistors of resistance 2.0 Ω are connected as shown.

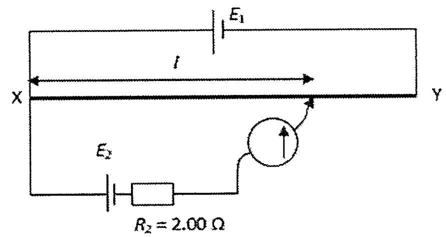


If the resistor R in the middle is removed from the circuit, what is the change in the effective resistance across **PQ**?

A Increase by 0.1 Ω **B** Increase by 0.4 Ω **C** Decrease by 0.4 Ω **D** Decrease by 2.0 Ω

28. [HCI/H2/Prelims 2010/P1/25]

A potentiometer consists of a 1.000 m long resistance wire XY in series with a battery of e.m.f. E_1 = 9.00 V and internal resistance r = 1.42 Ω. The resistance of XY is 8.30 Ω.

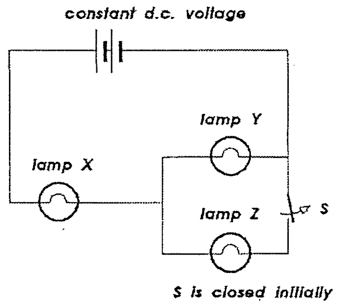


What is the e.m.f. of cell E_2 if the balance length l is found to be 0.745 m?

A 0.636 V **B** 2.72 V **C** 5.73 V **D** 9.00 V

29. [Dunman High/H2/Prelims 2010/P1/27]

In the circuit shown, X, Y and Z are three identical lamps. Initially switch S is closed.



When switch S is opened, the brightness of

A X decreases and that of Y increases.

B X increases and that of Y decreases.

C X stays the same and that of Y increases.

D X increases and that of Y stays the same.

30. [IJC/H2/Prelims 2010/P1/27]

When four identical lamps P, Q, R and S are connected as shown in diagram 1, they have normal brightness.

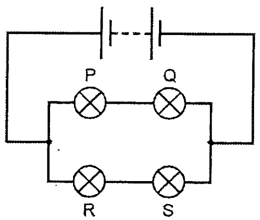


diagram 1

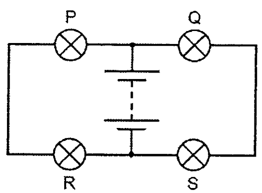


diagram 2

The four lamps and the battery are then connected as shown in diagram 2.

Which statement is correct?

A The lamps do not light up.

B The lamps are less bright than normal.

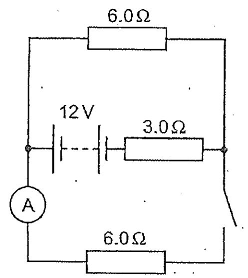
C The lamps have normal brightness.

D The lamps are brighter than normal.

31. [IJC/H2/Prelims 2010/P1/28]

In the circuit, the battery has an e.m.f. of 12 V and an internal resistance of 3.0 Ω. The ammeter has negligible resistance.

The switch is closed.

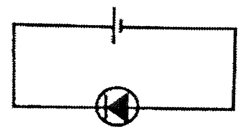


What is the reading on the ammeter?

A 0.50 A **B** 1.0 A **C** 1.3 A **D** 2.0 A

32. [MI/H2/Prelims 2010/P1/27]

A diode is connected to a battery as shown.



Which of the following statements about the diode is **not** true?

A No current flows through the diode.

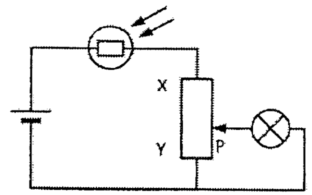
B The resistance is very large.

C The voltage across the diode is zero.

D The voltage across the diode is the same as the e.m.f. of the source.

33. [MI/H2/Prelims 2010/P1/28]

In the circuit shown, the light bulb will become brighter when



A light is incident on the LDR and **P** is moved to **X**.

B light is incident on the LDR and **P** is moved to **Y**.

C the LDR is covered and **P** is moved to **X**.

D the LDR is covered and **P** is moved to **Y**.

34. [MJC/H2/Prelims 2010/P1/27]

A row of 30 decorative lights, connected in series, is connected to a mains transformer. When the supply is switched on, the lights do not work. The owner uses a voltmeter to test the circuit. When the voltmeter is connected across the fifth bulb in the row, a reading of zero is obtained.

Which of the following scenarios described is **not** possible?

A Only the filament of the fifth bulb has broken.

B The fuse in the mains transformer has blown.

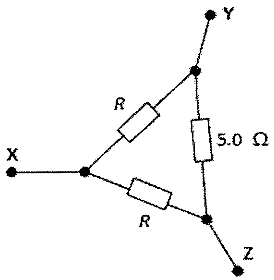
C The filament of at least one of the other bulbs has broken.

D There is a break in the wire from the supply to the transformer.

35. [MJC/H2/Prelims 2010/P1/28]

The diagram shows a network of three resistors. Two of these marked R, are identical. The other one has a resistance of 5.0 Ω.

The resistance between **Y** and **Z** is found to be 2.5 Ω.

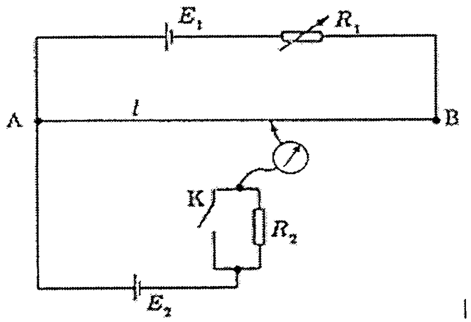


What is the resistance between **X** and **Y**?

A 1.00 Ω B 1.9 Ω C 2.5 Ω D 4.2 Ω

36. [NJC/H2/Prelims 2010/P1/29]

Which of the following statements about the circuit shown is **not** true?



A When switch K is closed, as the resistance R_2 increases, the balanced length l decreases.

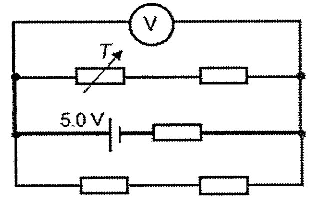
B When switch K is open, as the resistance R_2 increases, the balanced length l, remains unchanged.

C When switch K is open, as resistance R_1 increases, the balanced length l increases.

D Whether switch K is open or closed, changes in the internal resistance of E_2 will produce no change in the balanced length l.

37. [NYJC/H2/Prelims 2010/P1/27]

A cell of e.m.f. 5.0 V and negligible internal resistance is connected to four similar resistors and a variable resistor T, as shown.



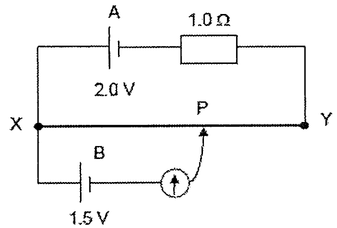
Resistance of each resistor is 1.0 kΩ and resistance of T is 5.0 kΩ.

What is the reading of the ideal voltmeter?

A 0 V B 2.0 V C 3.0 V D 5.0 V

38. [NYJC/H2/Prelims 2010/P1/28]

In the circuit as shown, cell A has an e.m.f. of 2.0 V and negligible internal resistance. Wire XY is 1.0 m long with a resistance of 4.0 Ω.



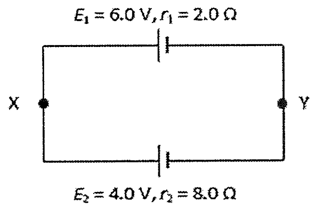
Cell B has an e.m.f. of 1.5 V and internal resistance 1.0 Ω.

What is the length XP required to produce null deflection in the galvanometer?

A 0.66 m B 0.75 m C 0.90 m D 0.94 m

39. [PJC/H2/Prelims 2010/P1/26]

The figure below shows a circuit with two batteries in opposition to each other. One has an e.m.f. E_1 of 6.0 V and internal resistance r_1 of 2.0 Ω and the other an e.m.f. E_2 of 4.0 V and internal resistance r_2 of 8.0 Ω.

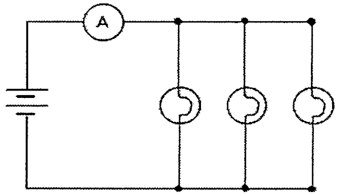


What is the potential difference across X and Y?

A 1.0 V B 2.0 V C 5.0 V D 5.6 V

40. [PJC/H2/Prelims 2010/P1/27]

Three similar light bulbs are connected to a constant voltage d.c. supply as shown in the diagram. Each bulb operates at normal brightness and the ammeter (of negligible resistance) registers a steady current.



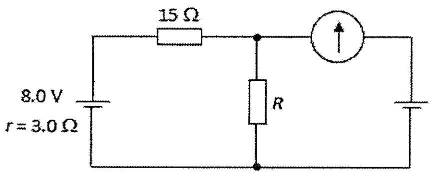
The filament of one of the bulb breaks.

What happens to the ammeter reading and to the brightness of the remaining bulbs?

	ammeter reading	bulb brightness
A	increases	decreases
B	increases	increases
C	unchanged	unchanged
D	decreases	unchanged

41. [RVHS/H2/Prelims 2010/P1/27]

In the circuit below, the 8.0 V cell has an internal resistance of 3.0 Ω. The galvanometer reading is zero.

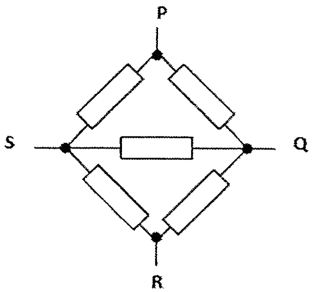


What is the resistance of R?

A 3.0 Ω **B** 5.0 Ω **C** 6.0 Ω **D** 9.0 Ω

42. [RI/H2/Prelims 2010/P1/27]

Five resistors of equal resistance are connected as shown.

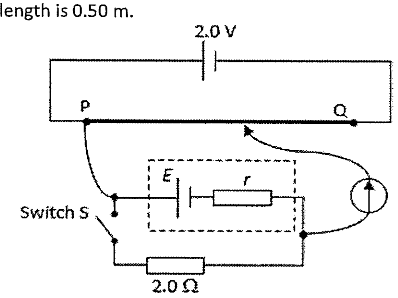


Which two points would give the maximum combination resistance?

A PQ **B** PR **C** PS **D** QS

43. [RI/H2/Prelims 2010/P1/28]

The diagram below shows a simple potentiometer circuit used to determine the internal resistance of a cell of e.m.f. E. The driver cell has an e.m.f. of 2.0 V with negligible internal resistance and the metre wire PQ is 1.0 m long. The cell is connected in parallel with a resistor of 2.0 Ω. When the switch is open, the balance length is 0.70 m and when the switch is closed, the balance length is 0.50 m.

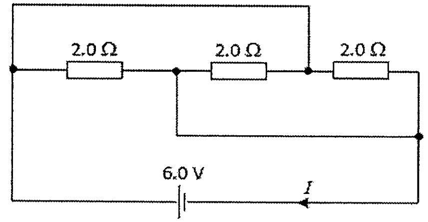


What is the internal resistance of the cell?

A 0.15 Ω **B** 0.40 Ω **C** 0.50 Ω **D** 0.80 Ω

44. [RVHS/H2/Prelims 2010/P1/26]

The battery in the circuit below has negligible internal resistance.



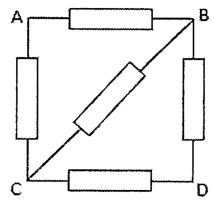
What is the current I?

A 1.0 A **B** 2.0 A **C** 4.5 A **D** 9.0 A

45. [SAJC/H2/Prelims 2010/P1/29]

The diagram shows a circuit consisting of five 1-Ω resistors. A multimeter is used to measure the resistance across different terminals.

Which of the following statements is **not** true?



A The resistance measured between B and C is 0.5 Ω.

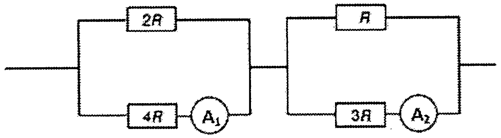
B The resistance measured between B and C will be smaller if an additional resistor is connected in parallel across AC.

C The resistance measured between A and D is 1 Ω.

D The resistance measured between A and D will be smaller if a zero resistance wire is connected across BC.

46. [TJC/H2/Prelims 2010/P1/27]

The ammeter A_1 of the circuit below reads 6.0 A.

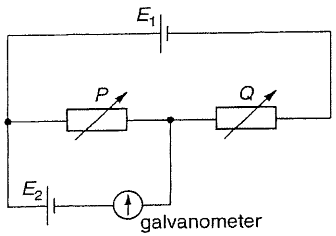


Assuming that both ammeters have negligible resistance, what is the reading on ammeter A_2?

A 4.5 A **B** 6.0 A **C** 13.5 A **D** 18.0 A

47. [TJC/H2/Prelims 2010/P1/26]

Two cells of e.m.f. E_1 and E_2 and negligible resistance are connected with two variable resistors as shown.



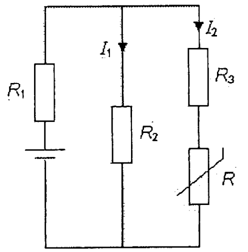
When the galvanometer deflection is zero, the resistances of the variable resistors are P and Q.

What is the value of the ratio $\frac{E_2}{E_1}$?

A $\frac{P}{Q}$ **B** $\frac{P}{P+Q}$ **C** $\frac{Q}{P+Q}$ **D** $\frac{P+Q}{P}$

48. [TJC/H2/Prelims 2010/P1/28]

In the circuit below, R_1, R_2 and R_3 are fixed resistors and R is a thermistor.



What happens to the currents I_1 and I_2 when the temperature at the thermistor increases?

A I_1 remains unchanged, I_2 increases.

B I_1 decreases, I_2 increases.

C I_1 increases, I_2 decreases.

D I_1 increases, I_2 increases.

49. [VJC/H2/Prelims 2010/P1/26]

A diode with the forward characteristic shown in Fig. 49.1 is connected in series with a variable, low voltage d.c. power supply, a meter of negligible internal resistance and a 50 Ω resistor as shown in Fig. 49.2.

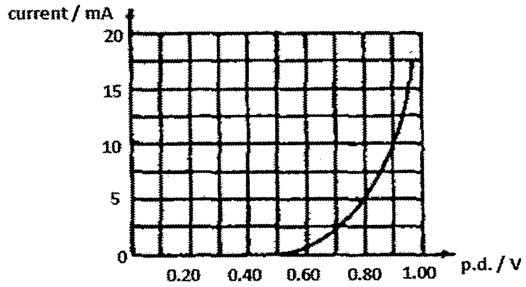


Fig. 49.1

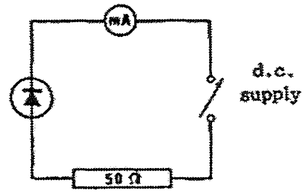


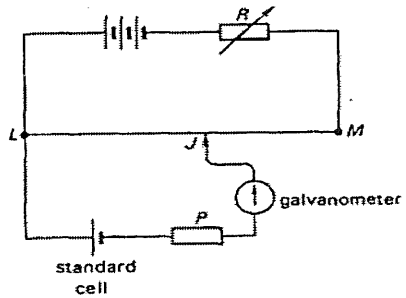
Fig. 49.2

When the meter reads 5 mA, what is the potential difference across the power supply?

A 0.25 V **B** 0.75 V **C** 1.05 V **D** 1.25 V

50. [VJC/H2/Prelims 2010/P1/27]

A potentiometer is to be calibrated with a standard cell using the circuit shown in the diagram.



The balance point is found to be near L. To improve accuracy the balance point should be nearer M. This may be achieved by

A Replacing the galvanometer with one of lower resistance

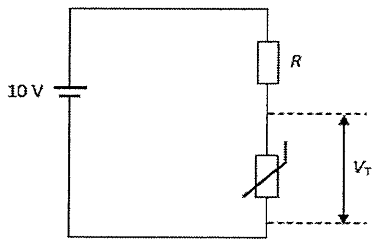
B Replacing the potentiometer wire with one of higher resistance per unit length

C Reducing the resistance R

D Increasing the resistance R

51. [YJC/H2/Prelims 2010/P1/27]

A thermistor is connected in series with a fixed resistor of resistance R and a cell of e.m.f. 10 V, as shown in the diagram.



When the temperature of the thermistor is 20°C, its resistance is 5.3 Ω and the potential difference V_T across it is 4.5 V.

What is the value of V_T if the temperature of thermistor increases to 60°C and the resistance drops to 3.1 Ω?

A 1.5 V **B** 2.6 V **C** 3.2 V **D** 3.5 V

II. Structured Questions

1. [CJC/H2/Prelims 2010/P2/5]

a. A set of coloured lamps are designed for use with a 240 V supply. The setup have 12 lamps connected as shown in Fig. 2.1**Error! Reference source not found.** below.

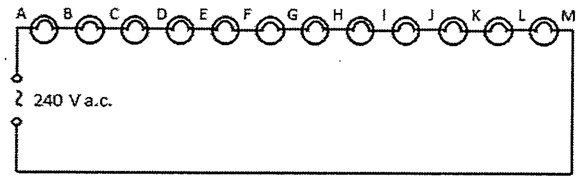


Fig. 2.1

However, the lamps do not light up when the set is plugged in. Therefore, a voltmeter is used to test the circuit. For each of the following observations, identify the fault.

i. The potential difference is zero across every lamp except **EF**, across which the potential difference is 240 V. **[1]**

ii. The potential difference between **A** and **M** is 240 V but the potential difference is zero across every lamp. **[1]**

b. i. Some lamps are designed so that when the filament fails the resistance of the lamp drops to zero. If this happens to one of the lamps in the above set up, calculate the fractional increase in the power dissipated in each of the remaining lamps, assuming that the resistance of these lamps does not change **[4]**

ii. What is likely to happen if failed lamps are not replaced? **[2]**

[Total: 8]

2. [AJC/H2/Prelims 2010/P3/7]

a. i. Define, in the context of an electrical circuit, the following

1. Coulomb **[1]**

2. Joule **[1]**

3. Watt. **[1]**

ii. Based on your answer in **a.i.**, derive the unit of potential difference in terms of SI base units. **[2]**

b. A car headlamp is marked 12 V, 72 W.

i. Calculate the working resistance of the car headlamp. **[2]**

ii. The lamp in **b.i.** is connected to a battery of e.m.f. of 12 V with an internal resistance of 0.20 Ω as shown in Fig. 1.1.

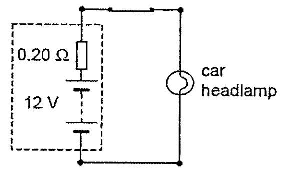


Fig. 1.1

The car headlamp is switched on for a 20-minute journey. Using your answer in **b.i.**, calculate

1. the current in the lamp **[2]**

2. the charge which passes through the lamp during the journey **[1]**

3. the energy supplied to the lamp during the journey. **[2]**

c. Two of the headlamps referred to in **b.** are connected in the circuit shown in Fig. 1.2, in which one source of e.m.f. (generator of the car) is placed in parallel with the car battery and the two lamps. Both lamps are on and are working normally.

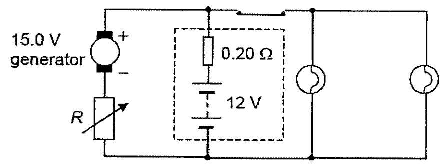


Fig. 1.2

The battery has an e.m.f. of 12.0 V and an internal resistance of 0.20 Ω. The generator has an e.m.f. of 15.0 V and negligible internal resistance. The generator is in series with a variable resistor R.

i. The value of R is adjusted such that there is no current in the battery when the lamps are on. Calculate

1. the current in the generator **[2]**

2. the value of the resistance of R. **[2]**

ii. Calculate the current in the battery when both lamps are switched off, the value of R remaining the same as in **i.**. **[2]**

d. Suggest two advantages which the circuit, as shown in **Error! Reference source not found.**Fig. 1.2 has over a single power source **[2]**

[Total: 20]

3. **[HCI/H2/Prelims 2010/P2/5]**

a. A student attempts to measure the resistivity of soil using two parallel copper plates driven into the ground as shown in Fig. 3.1.

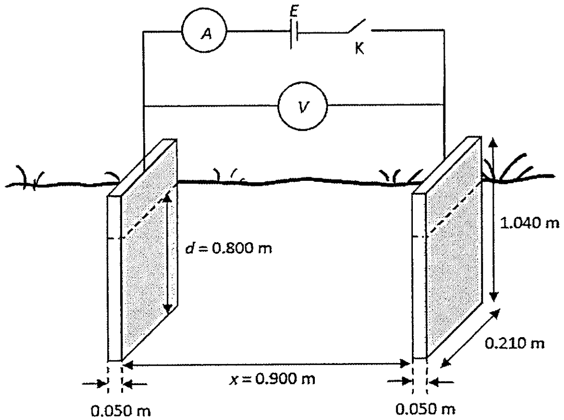


Fig. 3.1

Each copper plate has a height of 1.040 m, a width of 0.210 m and a thickness of 0.050 m. Assume the ammeter has zero resistance and the voltmeter has infinite resistance.

The copper plates are driven to a depth of d = 0.800 m and separated by a distance x = 0.900 m. If the soil is acidic, it reacts with copper and this produces an e.m.f. When switch K is open, the student obtained a steady voltmeter reading of +0.281 V. When switch K is closed, the student obtained a voltmeter reading of +1.398 V and an ammeter reading of 0.31 mA.

i. Show that the resistance of the soil between the copper electrodes is 3.6 kΩ. [2]

ii. Hence, find the resistivity of the soil. [2]

iii. Suggest how the value in **a.ii.** could be measured more accurately. [1]

b. A light bulb, marked 60 W, 120 V, is connected to an alternating power supply whose voltage V in volts is given by $V = (170\text{ V}) \times \sin\left[(100\pi\text{ s}^{-1})t\right]$ where t is time in seconds.

i. Calculate the value of the root-mean-square current in the bulb. [2]

ii. Sketch the graph showing the variation with time t of the power P dissipated by the light bulb. Indicate the mean power clearly on the graph. [2]

[Total: 9]

4. **[MJC/H2/Prelims 2010/P2/2 (part)]**

A photomultiplier tube is a device which has a common electrode (the photocathode) and a number of other electrodes (the dynodes), which must be maintained at definite potentials relative to the common electrode. A potential divider circuit may be used to supply these potentials.

Fig. 4.1 illustrates a photomultiplier tube with a photocathode and six dynodes, numbered 1 to 6. A potential divider of six resistors, each of resistance R, using a supply voltage of 1050 V, is connected to the electrodes in the tube.

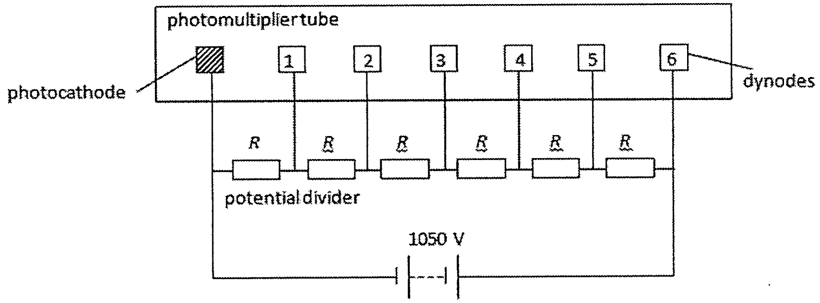


Fig. 4.1

a. For the situation where there is no electron current inside the photomultiplier tube, determine the potential difference between dynode 4 and the photocathode. [2]

b. A fault develops inside the photomultiplier tube causing a short circuit between dynodes 3 and 5. Determine the new potential difference between dynode 4 and the photocathode. Explain your reasoning. [3]

[Total: 5]

5. **[MJC/H2/Prelims 2010/P3/3]**

a. A thin layer of copper is deposited uniformly on the surface of an iron wire of radius 0.60 mm and length 3.0 m shown in Fig. 5.1.

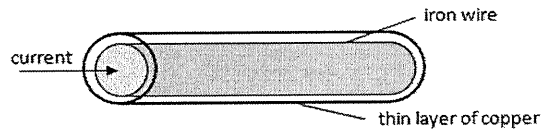


Fig. 5.1 (not to scale)

Determine the effective resistance between the ends of the copper-plated wire, given that the thickness of the copper is 1.78×10^{-5} m.

[Resistivity of iron = 8.90×10^{-8} Ω m; resistivity of copper = 1.60×10^{-8} Ω m] [3]

b. Fig. 5.2 shows a system in which an unmodulated audio frequency signal is transmitted from the transmitter to the receiver through a cable. The cable consists of two strands of insulated copper wire.

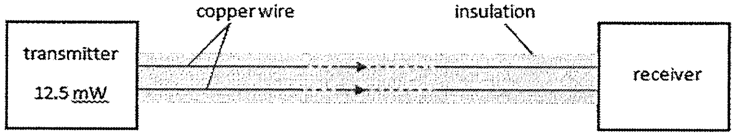


Fig. 5.2

The power output of the transmitter is 12.5 mW and the corresponding current in each wire is 2.5 mA. Power is lost to the surroundings due to the rise in temperature produced by this current. For transmitted signal to be detected the power input to the receiver must be at least 1.5 mW.

The resistance of each 1.0 m of the copper wire used in the cable is 0.27 Ω.

Calculate the maximum distance between the transmitter and receiver at which the transmission can be detected successfully. **[3]**

[Total: 6]

6. [RVHS/H2/Prelims 2010/P2/5]

a. Distinguish between *electromotive force* and *potential difference*. **[2]**

b. An electric hotplate is designed to operate on a power supply of 240 V has two coils of wire of resistivity of 9.8×10^{-7} Ω m. Each coil of wire has a length of 16 m of cross-sectional area 0.20 mm^2.

i. For one of the coils, calculate

1. its resistance **[2]**

2. the power dissipation when a 240 V supply is connected across it. **[2]**

ii. Fig. 6.1 shows how the two coils can be connected to operate at different powers.

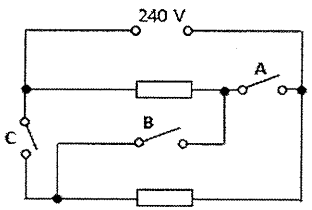


Fig. 6.1

On Fig. 6.2, fill up the table with "ON" or "OFF" to obtain the lowest and highest levels of operating power. **[2]**

	switch A	switch B	switch C
lowest			
highest			

Fig. 6.2

[Total: 8]

7. [VJC/H2/Prelims 2010/P3/4(modified)]

a. Define resistance. **[1]**

b. A wire with a resistance of 6.0 Ω is stretched so that its new length is three times its original length. Assuming that the resistivity and density of the material are not changed during the stretching process, calculate the resistance of the longer wire. **[3]**

c. The circuit in Fig. 7.1 is constructed of resistors, each of which has a maximum safe power rating of 0.40 W.

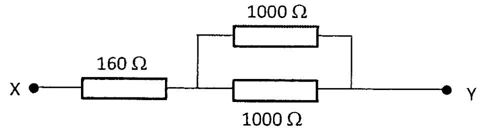


Fig. 7.1

i. Determine the maximum potential difference that can be applied between X and Y without damage to any of the resistors. **[3]**

ii. If this potential difference were exceeded, explain which resistor would be most likely to fail. **[2]**

[Total: 9]

8. [AJC/H2/Prelims 2010/P2/5]

Fig. 8.1 shows a potential divider circuit consisting of two resistors with resistances 10 kΩ and 40 kΩ respectively. The battery has an e.m.f. E and negligible internal resistance.

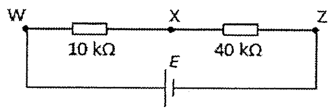


Fig. 8.1

a. Determine the potential difference V_{WX} across the 10 kΩ resistor in terms of E. **[2]**

b. A thermistor P is connected in parallel with the 10 kΩ resistor and a resistor of resistance 20 kΩ is connected in parallel with the 40 kΩ, as shown in Fig. 8.2.

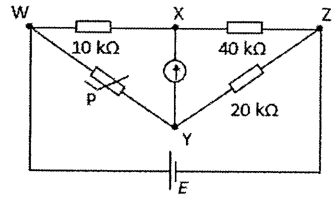


Fig. 8.2

A galvanometer is connected between X and Y. At room temperature of 29°C, it was found that there is no current flowing through the galvanometer.

i. Suggest what can be deduced about the potential at points X and Y at 29°C. **[1]**

ii. Determine the resistance of the thermistor P at 29°C. **[2]**

iii. The temperature of the thermistor is slowly raised.

1. On Fig. 8.2, indicate the direction of the current flowing between X and Y. **[1]**

2. Explain why a reading is detected on the galvanometer whenever the temperature is not at 29°C. **[1]**

[Total: 7]

9. [CJC/H2/Prelims 2010/P3/6 (part)]

The circuit in Fig. 9.1 is set up, with a 20 V driver cell and Cell P of 12 V, each with internal resistance 2.0 Ω.

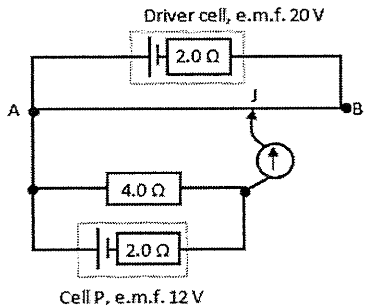


Fig. 9.1

The uniform resistance wire AB is of length 100 cm, and the balanced length AJ is found to be 64 cm.

a. Calculate the potential difference across the 4.0-Ω resistor. [2]

b. Determine the resistance of the wire AB. [3]

[Total: 5]

10. [Dunman High/H2/Prelims 2010/P3/4]

Jingwen wants to determine the internal resistance r of a cell of e.m.f. E using a potentiometer as shown in Fig. 10.1.

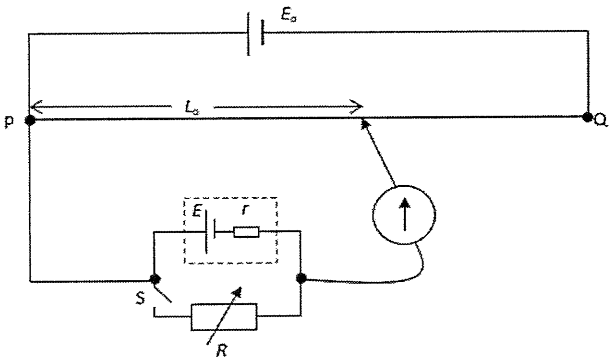


Fig. 10.1

When switch S is open, the balance length is L_o. When switch S is closed, the balance length is L.

a. i. When switch S is closed, obtain an expression for V, the potential difference across the variable resistor R in terms of the e.m.f. E and internal resistance r of the cell. [1]

ii. Hence or otherwise, show that the internal resistance r of the cell is

$$r = R\left(\frac{L_o}{L} - 1\right).$$ [2]

iii. She found that when R = 12 Ω, L_o = 82.3 cm and L = 78.9 cm. Determine r. [1]

b. She repeated the experiment with R set to **i.** 4.0 Ω and **ii.** 30 Ω.

Explain which value of R used would result in a more reliable determination of r. [2]

c. The value of r can also be found by drawing a suitable straight line graph. Explain how this can be done. [3]

[Total: 9]

11. [MI/H2/Prelims 2010/P3/3]

a. In Fig. 11.1, the battery has an internal resistance r and the ammeter has negligible resistance.

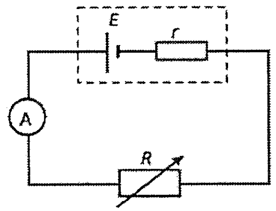


Fig. 11.1

Fig. 11.2 shows how current I in the circuit varies as the potential difference V across the variable resistor R changes.

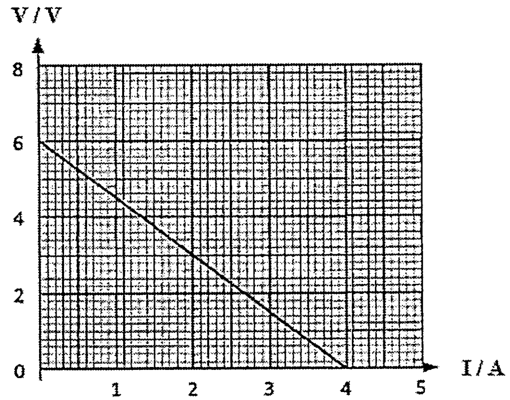


Fig. 11.2

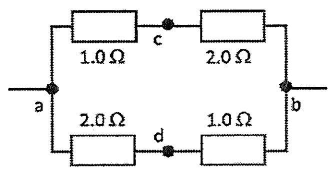


Fig. 10.3

i. Define potential difference. [2]

ii. 1. Show that the e.m.f E of the battery is 6.0 V. [2]

2. Calculate the power dissipated in the variable resistor R when the current in the circuit is 1.2 A. [2]

3. Calculate the internal resistance r of the cell. [2]

b. Four resistors are connected as shown in Fig. 10.3. Point **a** is at a higher potential than point **b**.

If a wire is connected from **c** to **d**, state and explain the direction of the current that will flow through the wire. [2]

[Total: 10]

12. **[NJC/H2/Prelims 2010/P3/4(part)]**

b. i. Fig. 12.1 shows a potentiometer setup where the potentiometer wire, ab, is not calibrated. E_s is a known standard cell. Describe how it is used to measure the e.m.f. of the unknown source E_x. **[2]**

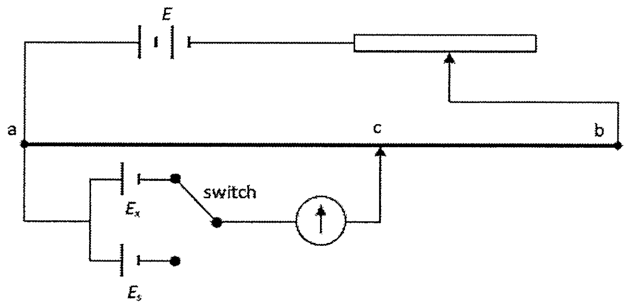


Fig. 12.1

ii. Discuss one advantage of using the potentiometer setup to measure the e.m.f. E_x. **[1]**

c. The potentiometer wire ab of length 1.00 m has a resistance of 600 Ω. The rheostat, R, has a resistance 400 Ω for the entire length of 50 cm. The previous circuit has been altered as shown in Fig. 12.2.

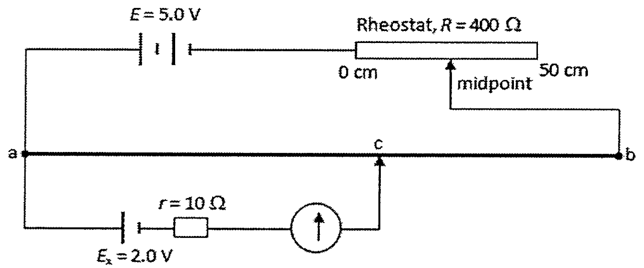


Fig. 12.2

i. Determine the balance length ac. **[2]**

ii. State the direction of the current flowing through the dry cell, E_x, when the rheostat R is adjusted from the midpoint to the right at the 40 cm mark. **[1]**

iii. Determine the new balance length, ac'. **[2]**

[Total: 8]

13. **[NYJC/H2/Prelims 2010/P3/3]**

Fig. 13.1. shows a circuit for measuring a small e.m.f. produced by a solar cell.

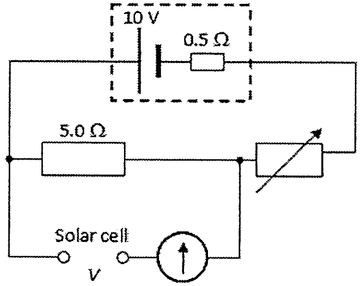


Fig. 13.1

a. The galvanometer shows null deflection when the variable resistor is set to 300 Ω. Determine the value of the e.m.f. V of the solar cell. **[2]**

b. Fig. 13.2 shows the 5.0 Ω resistor being replaced with a 1.2 m uniform resistance wire PQ of total resistance of 7.0 Ω. The variable resistor remains at 300 Ω.

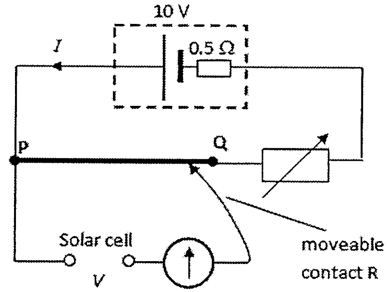


Fig. 13.2

i. Calculate the current I in the main circuit. **[2]**

ii. Calculate the distance from P that contact R must be connected to wire PQ such that the galvanometer shows null deflection. **[3]**

iii. Explain why this circuit is not suitable for measuring the e.m.f. of the solar cell when the value of the e.m.f. of the solar cell is of the order of millivolts. **[1]**

[Total: 8]

14. [PJC/H2/Prelims 2010/P3/3]

Fig. 14.1 shows a potential divider arrangement using a fixed resistor of resistance 4.0 kΩ and a variable resistor of maximum resistance 20 kΩ with a slide contact connected to terminal S.

The e.m.f. of the battery is 12 V and it has negligible internal resistance. It is possible to obtain different continuously-variable ranges by selecting, as the output, particular pairs of terminals from S, X, Y and Z.

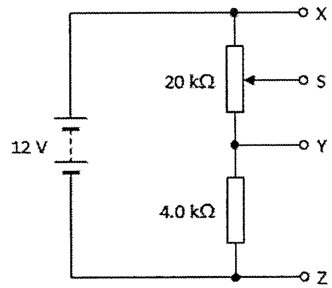


Fig. 14.1

a. i. Calculate the voltage range obtainable between the terminals S and X. [2]

ii. Hence, or otherwise, calculate the voltage range between the terminals S and Z. [1]

b. The slide contact S is set at the mid-point of the 20 kΩ resistance track. A voltmeter of resistance of 10 kΩ is then connected between S and Y. Calculate the reading on the voltmeter. [3]

c. The variable resistor in Fig. 14.1 is replaced by a thermistor T, as shown in Fig. 14.2. At room temperature, the resistance of the thermistor is 12 kΩ. When it is placed in hot liquid, its resistance falls to 2.0 kΩ.

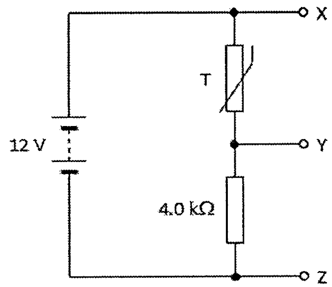


Fig. 14.2

On Fig. 14.3, sketch the temperature characteristic of the thermistor. [1]

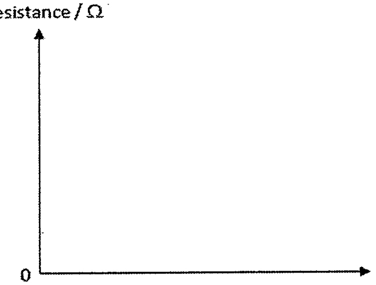


Fig. 14.3

[Total: 7]

15. [YJC/H2/Prelims 2010/P2/2]

Fig. 15.1. shows a potentiometer circuit that can be used to determine the unknown e.m.f. of a test cell. The driver cell has an e.m.f. of 12 V and internal resistance of 1.5 Ω. The resistance of the rheostat can vary between 0.0 Ω and 5.0 Ω and the resistance wire has a length of 1.2 m.

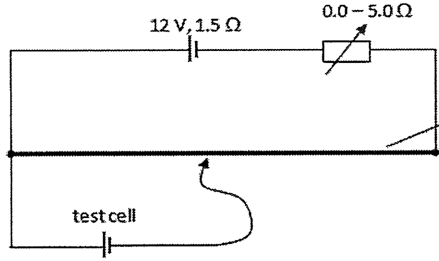


Fig. 15.1

a. When the resistance of rheostat is 2.3 Ω, the balance length is 0.57 m. When the resistance of rheostat is changed to 3.5 Ω, the balance length becomes 0.68 m. Calculate the e.m.f. of the test cell and the resistance of the 1.2 m long resistance wire. [4]

b. State what will happen to the balance length if the internal resistance of the test cell is doubled. [1]

c. Explain why the resistance of the rheostat cannot be higher than a particular value, if the potentiometer is to be able to determine the unknown e.m.f. [2]

[Total: 7]

16. [RI/H2/Prelims 2010/P3/2]

a. Distinguish between *resistance* and *resistivity* of a conductor. [2]

b. A cell of e.m.f. 2.50 V and internal resistance *R* is connected to two uniform resistive wires in series as shown in Fig. 16.1. The wires are made of the same material but have different lengths and diameters. Wire AB is 50.0 cm long and has a diameter *d*, whereas wire BC is 30.0 cm long and has a diameter 0.30 *d*. The ammeter and connecting wires are assumed to have no resistance.

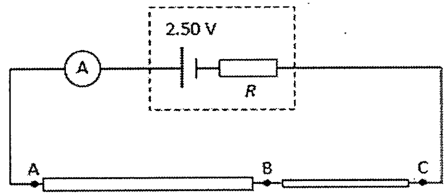


Fig. 16.1

Show that $\frac{R_{AB}}{R_{BC}} = 0.15$. [2]

c. A battery of e.m.f. 2.00 V and internal resistance r is connected across wire BC in parallel with another resistor of resistance r as shown in Fig. 16.2. The galvanometer shows no deflection when the jockey J is at the midpoint of wire BC.

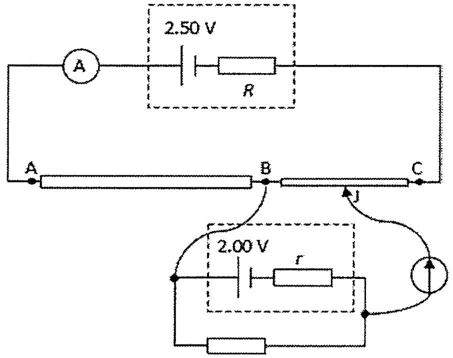


Fig. 16.2

i. Show that V_{BC} = 2.00 V. [1]

ii. Determine the internal resistance R of the 2.50 V cell if the ammeter shows a reading of 0.400 A. [3]

d. Suggest and explain whether your answer in part **c.ii.** is an over-estimate or under-estimate if the ammeter is not ideal. [2]

[Total: 10]

17. [SAJC/H2/Prelims 2010/P3/4]

a. Fig. 17.1 shows a cell of e.m.f. 2.0 V and internal resistance 0.20 Ω connected in parallel to two identical lamps L_1 and L_2. The ammeters A_1 and A_2 in the circuit have negligible resistance and A_2 reads 0.50 A.

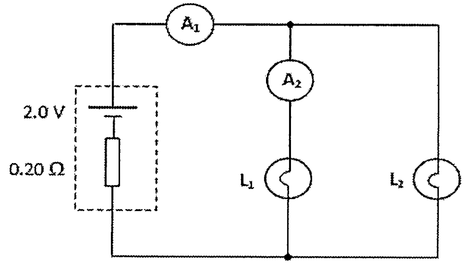


Fig. 17.1

i. Calculate the potential difference across L_1. [2]

ii. If another identical lamp L_3 is connected in parallel with L_1 and L_2, explain whether the current in ammeter A_1 remains the same, increases or decreases. [2]

b. Fig. 17.2 shows a circuit which is used to measure the e.m.f. of Cell **Y**.

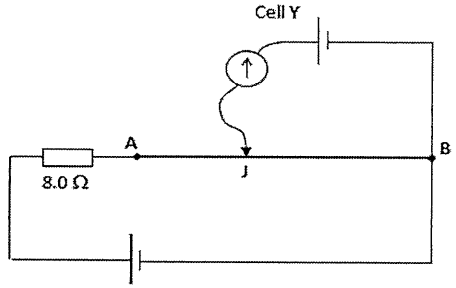


Fig. 17.2

Cell **X** has an e.m.f. of 2.00 V and negligible internal resistance. It is connected in series with an 8.0 Ω resistor and resistance wire **AB**. The resistance wire **AB** has a length 100.0 cm and a resistance 2.0 Ω.

i. Calculate the potential difference across **AB**. [1]

ii. The movable contact **J** is now moved along **AB**. When the galvanometer indicates a zero reading, the length **AJ** is 30.0 cm. Calculate the e.m.f., in mV, of Cell **Y**. [2]

iii. Without changing the length **AB**, suggest three modifications to the circuit in Fig. 17.2 that would cause the contact **J** to be closer to **A** when the galvanometer gives a zero reading. [3]

[Total: 10]

	Raffles Institution Physics Department Revision Package 1	**Circuits**	**16**

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	A	C	A	B	D	B	D	B	C	A	C	A	C	A

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
B	B	D	C	C	A	C	D	D	A	B	A	C	A	C

31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
B	C	A	A	B	A	C	D	D	D	C	B	D	D	D

46	47	48	49	50	51
A	B	B	C	D	C

1 B

$$R_{total} = \frac{V}{I} = \frac{12}{3.0} = 4.0\,\Omega$$

$$R = 4.0 - 0.5 = 3.5\,\Omega$$

With ideal ammeter (i.e. zero resistance),

$$I_{ideal} = \frac{V}{I} = \frac{12}{3.5} = 3.429\text{ A}$$

$$\therefore \%\,\text{error} = \frac{3.429 - 3.0}{3.429} \times 100\% = 13\%$$

2 A

$$R = \rho\frac{L}{A} = \rho\frac{L}{\pi r^2}$$

Given same length and same material, $R \propto \frac{1}{d^2}$ $\quad \therefore R_P = 4R_Q$

P and Q are parallel, $I_P R_P = I_Q R_Q$

$$\frac{I_P}{I_Q} = \frac{R_Q}{R_P} = \frac{R_Q}{4R_Q} = \frac{1}{4}$$

$$\frac{I_P}{I_{total}} = \frac{1}{5} = 0.20$$

3 C

$$V = AL \;\therefore L_{Al} = \frac{1}{3}L_{Cu}$$

$$R = \rho\frac{L}{A}$$

$$\frac{R_{Al}}{R_{Cu}} = \frac{\rho_{Al}}{\rho_{Cu}}\left(\frac{L_{Al}}{L_{Cu}}\right)\left(\frac{A_{Cu}}{A_{Al}}\right) = 3\left(\frac{1}{3}\right)\left(\frac{1}{3}\right) = \frac{1}{3}$$

4 A Refer to lecture notes' definition.

5 B Current is the same for resistors in series, for P at 10 V, current is 2A → p.d. across Q is 5 V. Total p.d. = 15 V.

6 D Assume NTC thermistor, when temperature increases, resistance decreases → lower p.d. (A wrong)
Removing earth connection at M does not change p.d. across thermistor (B wrong)
Closing switch S reduces equivalent resistance of // arrangement of thermistor and R_2 → lower p.d. (C wrong)
Reduce R_3 → p.d. across thermistor increases.

7 B

$$R = \rho\frac{L}{A} = \rho\frac{L}{\pi d^2/4}$$

Given same resistance and same length,

$$R_{Al} = R_{Ag} = R$$

$$\rho_{Al}\frac{4L}{\pi d_{Al}^{\,2}} = \rho_{Ag}\frac{4L}{\pi d_{Ag}^{\,2}}$$

$$\frac{d_{Ag}^{\,2}}{d_{Al}^{\,2}} = \frac{\rho_{Ag}}{\rho_{Al}} = \frac{1}{2}$$

$$d_{Ag} = \sqrt{0.5}\,d_{Al} = 0.71\,d$$

8 D Since 2nd bulb filament is broken, voltmeter X is now directly connected to the power source and p.d. across X is 240 V.
There is no current in the circuit and p.d. across the other five lightbulbs is 0 V.

9 B From B to C, there is a drop of 0.25 V. This is p.d. lost to internal resistance.

$$E - V_T = Ir$$

$$0.25 = \left(\frac{E}{R+r}\right)r$$

$$0.25 = \left(\frac{3.5(0.50)}{14+r}\right)r \qquad \text{(from CRO screen: } E = 3.5 \text{ div x } 0.50 \text{ V div}^{-1}\text{)}$$

$$3.5 + 0.25r = 1.75r$$

$$r = 2.3\ \Omega$$

10 C

Combined resistance of R_1 and $R_3 = \frac{\rho l}{A} + \frac{0.5\rho(2l)}{A} = \frac{2\rho l}{A}$

This is parallel with R_2,

$$\frac{1}{R_{XY}} = \frac{1}{\frac{2\rho(2l)}{A}} + \frac{1}{\frac{2\rho l}{A}} = \frac{3A}{4\rho l} \Rightarrow R_{XY} = \frac{4\rho l}{3A} = 1.3\frac{\rho l}{A}$$

11 A Only bulb 3 gets brighter. When bulb 1 filament breaks, it is as good as removing bulb 1, leaving only bulb 3 there. Hence equivalent resistance there is now the resistance of bulb 3 (increase). This results in an increase in p.d. across bulb 3 and a decrease in p.d. for rest of the bulbs/ arrangement in series with bulb 3. (No calculations required)

12 C

$$R = \frac{\rho L}{A} = \frac{\rho L}{\pi d^2/4} = 2.000\ \text{k}\Omega$$

$$R' = \frac{\rho(1.004L)}{\pi(0.99d)^2/4} = 1.024\frac{\rho L}{\pi d^2/4} = 1.024(2.000) = 2.049\ \text{k}\Omega$$

13 A

$$R = \frac{\rho L}{A} = \frac{\rho L}{\pi(d^2/4)}$$

$$R_{Cu} = \frac{1.7\times10^{-8}(10)}{\pi\left(\frac{0.0050^2}{4}\right)} = 8.658\times10^{-3}\ \Omega$$

$$R_{Al} = \frac{2.7\times10^{-8}(10)}{\pi\left(\frac{0.010^2}{4} - \frac{0.0050^2}{4}\right)} = 4.584\times10^{-3}\ \Omega$$

Parallel arrangement,

$$\frac{1}{R} = \frac{1}{R_{Cu}} + \frac{1}{R_{Al}} = \frac{1}{8.658\times10^{-3}} + \frac{1}{4.584\times10^{-3}}$$

$$R = 0.002997 = 0.0030\ \Omega$$

14 C

$$10 = \frac{4.0}{4.0 + 0.02 + R_{wires}} \times 12$$

$$4.0 + 0.02 + R_{wires} = 4.8$$

$$R_{wires} = 0.78\ \Omega$$

$$V_{wires} = \frac{0.78}{4.8} \times 12 = 1.95\ \text{V}$$

$$P_{wires} = \frac{V_{wires}^2}{R_{wires}} = \frac{1.95^2}{0.78} = 4.875 = 4.9\ \text{W}$$

15 A

$$R = \rho\frac{L}{A} = \rho\frac{L}{L^2} = \frac{\rho}{L}$$

$$\frac{R_X}{R_Y} = \frac{\frac{\rho}{2L}}{\frac{\rho}{L}} = \frac{1}{2}$$

16 B Initially, graph shows I is proportional to V, or vice versa. This implies constant resistance.

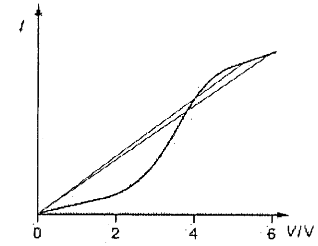
Next, I increases at a slower rate w.r.t. V, this implies resistance ($R = V/I$) is increasing. Hence graph B is correct.

17 B Initially, graph shows I is proportional to $V \rightarrow$ constant R

Then, I increases at an increasing rate w.r.t. $V \rightarrow$ R decreases

Later, I increases at a decreasing rate w.r.t. $V \rightarrow$ R increases

Finally, there is a constant slope $\rightarrow$ but this does not mean constant R (drawing straight lines from origin shows decreasing gradient lines $\rightarrow$ ratio of V to I increases, i.e. R increases). Graph B is correct.



18 D

$$E = V_{total}$$

$$= \frac{W_{total}}{Q} \qquad \text{(V is electrical energy per unit charge converted to other forms of energy)}$$

$$= \frac{88.3 + 45.2}{30}$$

$$= 4.45\ \text{V}$$

19 C Note vertical axis starts from 10 mA. Hence V is not proportional to I, thus it does not obey Ohm's Law.

Resistance at 2.0 V = $V / I = 2.0 / (50.0 \times 10^{-3}) = 40.0\ \Omega$

20 C Adding another bulb in parallel with bulb 2 (and bulb 3) will reduce the equivalent resistance for the parallel arrangement. This results in higher p.d. across bulb 1 and bulb 4.

Removing bulb 2 *increases* equivalent resistance, not reduces it (B wrong)

A and D result in lower p.d. across bulb 1.

21 A Since the 2.0 Ω resistor and resistor R is in parallel, potential difference across each of the resistor is the same.

$V_{2.0\Omega} = V_R$

$I_2(2.0) = 2.0(R)$

$I_2 = \dfrac{2.0(1.0)}{2.0} = 1.0 \text{ A}$

Consider the full circuit and assuming $V_{ab} < 12$ V,

$12 - V_{ab} = 3.0(1.0) + 2.0 + 2.0(3.0)$

$V_{ab} = 1.0 \text{ V}$

Alternative solution:

Consider the full circuit and assuming $V_{ab} > 12$ V,

$V_{ab} - 12 = 3.0(1.0) + 2.0 + 2.0(3.0)$

$V_{ab} = 23.0 \text{ V}$

22 C

$I_{bulb} = \dfrac{P}{V} = \dfrac{2.5}{3.0} = 0.83 \text{ A}$

$V_R = RI = V_{batt} - 3.0$

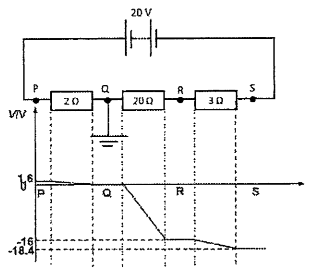
$R = \dfrac{9.0 - 3.0}{0.83} = 7.2 \ \Omega$

23 D

$V_{PQ} = \dfrac{2}{2+20+3}(20) = 1.6 \text{ V}$

$V_{QR} = \dfrac{20}{25}(20) = 16 \text{ V}$

$V_{RS} = \dfrac{3}{25}(20) = 2.4 \text{ V}$



24 D

$E = I_1(R_1 + r) = 3.0(1.0 + r)$

$E = I_2(R_2 + r) = 2.0(2.0 + r)$

Solving, $r = 1$, $E = 6.0$ V

25 A Since the resistors are in parallel, the equivalent resistance will be smaller than the individual resistances. Thus we will have the maximum current, resulting in the maximum power.

The circuit with the maximum resistance, which is S2 will give minimum power.

26 B

From $R_{eq} = \dfrac{R_2 R_1}{R_2 + R_1} = \dfrac{R_1}{1 + R_1/R_2}$

As R_2 increases, R_{eq} approaches R_1.

27 A For 5 resistors, $R_{5R} = \dfrac{R}{5}$, For 4 resistors, $R_{4R} = \dfrac{R}{4}$

$\Delta R_{eff} = \dfrac{R}{4} - \dfrac{R}{5} = \dfrac{1}{20}(2.0) = 0.1 \ \Omega$

28 C

$V_{XY} = \dfrac{R_{XY}}{R_{XY} + r}E_1 = \dfrac{8.30}{8.30 + 1.42}(9.00) = 7.69 \text{ V}$

$E_2 = \dfrac{L_{XJ}}{L_{XY}}V_{XY} = \dfrac{0.745}{1.000}(7.69) = 5.73 \text{ V}$

29 A Let R be the resistance of the 3 identical bulbs and ε the constant d.c. voltage.

When S is closed:

$I = \dfrac{\varepsilon}{R + \frac{R}{2}} = \dfrac{2}{3}\dfrac{\varepsilon}{R}$

$P_{lampX} = I^2R = \left(\dfrac{2}{3}\dfrac{\varepsilon}{R}\right)^2 R = 0.44\dfrac{\varepsilon^2}{R}$

$P_{lampY} = I^2R = \left(\dfrac{1}{2} \times \dfrac{2}{3}\dfrac{\varepsilon}{R}\right)^2 R = 0.11\dfrac{\varepsilon^2}{R}$

When S is opened:

$I = \dfrac{\varepsilon}{R + R} = \dfrac{1}{2}\dfrac{\varepsilon}{R}$

$P_{lampX} = P_{lampY} = I^2R = \left(\dfrac{1}{2}\dfrac{\varepsilon}{R}\right)^2 R = 0.25\dfrac{\varepsilon^2}{R}$

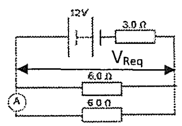
30 C Diagram 1 is the equivalent circuit of Diagram 2 since all lamps are identical.

31 B

$R_{eq} = \dfrac{6.0(6.0)}{6.0 + 6.0} = 3.0 \ \Omega$

$V_{R_{eq}} = \dfrac{3.0}{3.0 + 3.0}(12) = 6.0 \text{ V}$

$I = \dfrac{V_{R_{eq}}}{R} = \dfrac{6.0}{6.0} = 1.0 \text{ A}$



32 C The voltage across the diode is the emf of the battery assuming no internal resistance.

33 A The bulb becomes brighter when R_{XY} increases. This occurs when:

1. Light intensity increases, R_{LDR} decreases.
2. Shifting P to X to increase the resistance of the variable resistor R_{XY}.

34 A If the filament of the fifth bulb has broken, the voltmeter will register the potential difference across the transformer and not a zero reading.

35 B
$$R_{YZ} = \frac{5.0(2R)}{5.0 + 2R} = 2.5\ \Omega$$
$$R = 2.5\ \Omega$$
$$R_{XY} = \frac{2.5(7.5)}{2.5 + 7.5} = 1.9\ \Omega$$

36 A Since there is no current passing through the secondary circuit, even when resistance R_2 increases, the balanced length is unaffected.

37 C
$$R_{eq} = \frac{6.0(2.0)}{6.0 + 2.0} = 1.5\ \text{k}\Omega$$
$$V = \frac{1.5}{1.5 + 1.0}(5.0) = 3.0\ \text{V}$$

38 D
$$V_{XY} = \frac{R_{XY}}{R_{XY} + r}E_1 = \frac{4.0}{5.0}(2.0) = 1.6\ \text{V}$$
$$E_2 = \frac{L_{XP}}{L_{XY}}V_{XY} = 1.5$$
$$L_{XP} = \frac{L_{XY}}{V_{XY}}1.5 = 0.94\ \text{m}$$

39 D
$$E_{eff} = E_1 - E_2 = 2.0\ \text{V}$$
$$I = \frac{2.0}{8 + 2} = 0.2\ \text{A}$$
$$V_{XY} = 6.0 - (0.2)(2) = 5.6\ \text{V}$$

40 D For 3 identical bulbs in parallel: $R_{eq} = \frac{R}{3}$. For 2 identical bulbs in parallel: $R_{eq} = \frac{R}{2}$.

When the filament of a bulb breaks, R_{eq} increases and hence I will decrease.

41 C
$$V_R = 2.0 = \frac{R}{R + 15 + 3} \times 8.0 \text{ (this is a potentiometer arrangement)}$$
$$R = 6.0\ \Omega$$

42 B

Across PQ and PS: $R_{eff} = \frac{5}{8}R$

Across PR: $R_{eff} = R$

Across QS: $R_{eff} = \frac{1}{2}R$

43 D When switch is opened, $E = \frac{L_{PJ}}{L_{PQ}}V_{PQ} = \frac{0.7}{1.0}(2.0) = 1.4\ \text{V}$

When switch is closed, $V_{PJ} = \frac{0.5}{1.0}(2.0) = \frac{R}{R + r}E = \frac{2.0}{2.0 + r}(1.4) \Rightarrow r = 0.80\ \Omega$

44 D
$$R_{eq} = \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2}\right)^{-1} = \frac{2}{3}\ \Omega$$
$$I = \frac{V}{R_{eq}} = \frac{6.0}{\frac{2}{3}} = 9.0\ \text{A}$$

45 D Since the circuit is symmetrical about BC, there is no current passing through BC as B and C are at the same potential.

46 A For circuits in parallel, potential across is the same.
$$V_{2R} = V_{4R}$$
$$I_{2R}(2R) = (6.0)(4R)$$
$$I_{2R} = 12\ \text{A}$$
$$I_0 = I_{2R} + I_{4R} = 18\ \text{A}$$

Effective resistance of R and 3R resistors in parallel $= \left(\frac{1}{R} + \frac{1}{3R}\right)^{-1} = 0.75R$

p.d. across R and 3R resistors in parallel = p.d. across 3R resistor
$$18 \times 0.75R = I_{3R}(3R)$$
$$I_{3R} = 4.5\ \text{A}$$

47 B Since no current passes through the galvanometer, $E_2 = \frac{P}{P + Q}E_1 \Rightarrow \frac{E_2}{E_1} = \frac{P}{P + Q}$

48 B When $T_{themistor}$ increases, $R_{thermistor}$ decreases. Overall resistance decreases, hence, total current I_0 increases. Hence, p.d. across R_2 ($E - I_0R_1$) decreases. Hence, I_1 decreases. Since $I_0 = I_1 + I_2$, if I_0 increases and I_1 decreases, I_2 must increase.

49 C At 5 mA, potential across the diode = 0.8 V
$$V_{DCSupply} = 0.8 + 50(5 \times 10^{-3}) = 1.05\ \text{V}$$

50 D
$$V_{LM} = \frac{R_{LM}}{R_{LM} + R}E_1$$
$$E_{\text{Standardcell}} = \frac{L_{LJ}}{L_{LM}}V_{LM}$$

For L_U to increase, V_{LM} must decrease for the same $E_{standardcell}$. Increasing R will decrease the fraction $\dfrac{R_{LM}}{R_{LM}+R}$ and hence decrease V_{LM}.

51 C At 20°C, $V_T=\dfrac{R_T}{R_T+R}E=\dfrac{5.3}{5.3+R}(10)=4.5 \Rightarrow R=6.5\ \Omega$

At 60°C, $V_T=\dfrac{3.1}{3.1+6.5}10=3.2$ V

Structured Questions

1. **[CJC/H2/Prelims 2010/P2/5]**

a. i. The filament of the lamp between **E** and **F** fails.

ii. One or more connecting wires between the lamps are damaged.

b. i. Let R be the resistance of one lamp, E the e.m.f. provided by the power supplied, and I the current in the circuit.

Before the failure of one lamp, we have: $R_{eff}=12R \Rightarrow I=\dfrac{V}{12R}$.

Thus, power dissipated in one lamp: $P=I^2R=\dfrac{V^2}{(12R)^2}\times R=\dfrac{V^2}{144R}$

After the failure of one lamp, we have: $R_{eff}'=11R \Rightarrow I'=\dfrac{V}{11R}$.

Thus, power dissipated in one lamp: $P'=(I')^2R=\dfrac{V^2}{(11R)^2}\times R=\dfrac{V^2}{121R}$

Fractional increase in the power dissipated: $\dfrac{\Delta P}{P}=\dfrac{P'-P}{P}=\dfrac{P'}{P}-1=\dfrac{\frac{V^2}{121R}}{\frac{V^2}{144R}}-1=\dfrac{144}{121}-1=0.190$

ii. The durability of the remaining lamps will decrease, since the power dissipated as heat in each of them is higher than normal.

2. **[AJC/H2/Prelims 2010/P3/7]**

a. i. 1. One coulomb is defined as the amount of electric charge that passes through a point in a circuit in one second when there is a constant current of one ampere.

2. 1 joule is the work required to move an electric charge of one coulomb through an electrical potential difference of one volt.

3. 1 watt is the rate at which work is done when one ampere of current flows through an electrical potential difference of one volt.

ii. $V=\dfrac{W}{Q} \Rightarrow$ unit of $V=\dfrac{\text{unit of } W}{\text{unit of } Q}=\dfrac{\text{kg m s}^{-2}\times\text{m}}{\text{A s}}=\text{kg m}^2\ \text{A}^{-1}\ \text{s}^{-3}$

b. i. $P=\dfrac{V^2}{R} \Rightarrow R=\dfrac{V^2}{P}=\dfrac{(12)^2}{72}=2.0\ \Omega$

ii. 1. $I=\dfrac{V}{R+r}=\dfrac{12\text{ V}}{2.0+0.20}=\dfrac{12}{2.2}=5.45$ A

2. $Q=It=(5.45)(20\times60)=6550$ C

3. Energy supplied to the lamp = heat dissipated at the lamp.

$E=I^2Rt=(5.45)^2\times(2.0)\times(20\times60)=71.3$ kJ

c. i. 1. If there is no current in the battery, the p.d. across the battery and the lamps must be equal to the e.m.f. of the battery, i.e. 12 V.

At a p.d. of 12 V, current in each lamp: $I_{lamp}=\dfrac{12}{2.0}=6.0$ A

Hence, current in the generator is: $I_{generator}=2I_{lamp}=2(6.0)=12$ A

2. $V_{lamp}=E_{generator}-IR \Rightarrow R=\dfrac{E_{generator}-V_{lamp}}{I}=\dfrac{15.0-12.0}{12}=0.25\ \Omega$

ii. When the lamps are switched off, the generator, the resistor and the battery are in series. Thus, the net e.m.f. of the circuit is: $E_{net}=15.0-12.0=3.0$ V

Current in battery: $I=\dfrac{E_{net}}{R+r}=\dfrac{3}{0.25+0.20}=6.7$ A

d. 1. Power is available to the lamps as long as one of the e.m.f. sources is available.
2. When the lamps are switched off, the power from the generator could be used to charge the car's battery.

3. **[HCI/H2/Prelims 2010/P2/5]**

a. i. When the switch is open, the voltmeter measures the e.m.f. of the chemical cell alone.

$E_{chemical}=+0.281$ V

When the switch is closed, the voltmeter measures the e.m.f. of the external cell.

$E_{external}=+1.398$ V

When the switch is closed, the two cells are connected in series. Since both e.m.f. values measured are positive, the terminals of the same sign of the two cells are connect to each other (i.e. positive to positive).

$E_{net}=E_{external}-E_{chemical}=+1.398-0.281=1.117$ V

Resistance of the soil between the copper electrodes: $R=\dfrac{E_{net}}{I}=\dfrac{1.017}{0.31\times10^{-3}}=3.60\ \text{k}\Omega$

ii. $R=\rho\dfrac{L}{A} \Rightarrow \rho=\dfrac{RA}{L}=\dfrac{3603\times(0.210\times0.800)}{0.900}=673\ \Omega$ m

iii. Any one

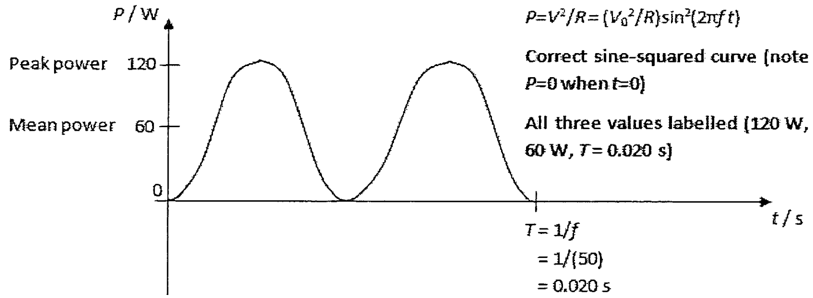
- Among all the readings given, the least significant or most imprecise is the current reading (only 2 s.f.). The current reading will be subject to significant random errors.
  - EITHER increase the area of the copper plates in the soil. This will decrease the resistance of the sample of soil to be measured and increase the current readings for the same voltage applied.
  - OR use a higher voltage supply so that current will be greater.
- Use a variable voltage supply in place of the battery. Plot the graph of the voltmeter reading V against the current I. The gradient equals the resistance R of the soil (and the y-intercept is the e.m.f. due to the copper plate interacting with the soil). Calculate the resistivity using the value of resistance R obtained from the gradient of the graph.

- Vary the area of the copper plates in the soil (use different depths), find the corresponding R of the soil between the plates using V and I, and plot the graph of the R against A^{-1}. The gradient is ρL where $L = x$. Calculate resistivity ρ as gradient divided by x.

b. **i.** $V_{r.m.s.} = \dfrac{V_0}{\sqrt{2}} = \dfrac{170}{\sqrt{2}} = 120.20$ V

$\langle P \rangle = I_{r.m.s.} V_{r.m.s.} \Rightarrow I_{r.m.s.} = \dfrac{\langle P \rangle}{V_{r.m.s.}} = \dfrac{60}{120.20} = 0.499$ A

ii. Compare the given equation with $V = V_0 \sin(\omega t) = V_0 \sin(2\pi f t)$, we have $100 = 2f \Rightarrow f = 50\ s^{-1}$



4. **[MJC/H2/Prelims 2010/P2/2 (part)]**

a. Although there is no current inside the tube, there is current flowing through the external resistors.

The current through the 6 resistors: $I = \dfrac{1050}{6R}$ A

The potential difference between dynode 4 and the photocathode:

$$V_{\text{photocathode}-4} = IR_{\text{photocathode}-4} = \frac{1050}{6R} \times 4R = 700 \text{ V}$$

OR Using potential divider rule, p.d. across photocathode and dynode 4 $= \dfrac{4R}{6R} \times (1050) = 700$ V

b. With dynode 3 and 5 short-circuited, the resistance between them is zero. Current will flow directly from dynode 5 to dynode 3, bypassing the two resistances between dynode 3 and 5 completely.

Since there is no current flowing through the two resistors between dynode 3 and dynode 5, there is no potential difference across either of them. Hence, the potential at dynode 3 is equal to potential at dynode 4, which is also equal to the potential at dynode 5.

The total resistance of the entire series dropped from $6R$ to $4R$.

New current: $I' = \dfrac{1050}{4R}$ A

The potential difference between dynode 3 and the photocathode is:

$$V_{\text{photocathode}-3} = IR_{\text{photocathode}-3} = \frac{1050}{4R} \times 3R = 788 \text{ V}$$

5. **[MJC/H2/Prelims 2010/P3/3]**

a. for the iron wire: $R_{\text{iron}} = \dfrac{\rho_{\text{iron}} l}{A_{\text{iron}}} = \dfrac{(8.90 \times 10^{-8}) \times (3.0)}{\pi \times (0.60 \times 10^{-3})^2} = 0.23608\ \Omega$

for the copper layer: $R_{\text{copper}} = \dfrac{\rho_{\text{copper}} l}{A_{\text{copper}}} = \dfrac{(1.60 \times 10^{-8}) \times (3.0)}{\pi \times \left[(0.60 \times 10^{-3} + 1.78 \times 10^{-5})^2 - (0.60 \times 10^{-3})^2\right]} \approx 0.70485\ \Omega$

When copper is deposited on the surface of the wire, it acts as a parallel shunt to the iron wire.

$$\frac{1}{R_{\text{eff}}} = \frac{1}{R_{\text{iron}}} + \frac{1}{R_{\text{copper}}}$$

$$R_{\text{eff}} = \left(\frac{1}{0.23608} + \frac{1}{0.70485}\right)^{-1} = 0.177\ \Omega$$

b. **Method 1:**

Power loss per metre of cable (comprising 2 wires) $= 2 \times I^2 R = 2 \times (2.5 \times 10^{-3})^2 \times (0.27) = 3.375 \times 10^{-6}$ W

Maximum power loss in the cable $= 12.5 - 1.5 = 11.0$ mW $= 11.0 \times 10^{-3}$ W

Hence, maximum distance $= \dfrac{11.0 \times 10^{-3}}{3.375 \times 10^{-6}} = 3260$ m

Method 2:

Maximum power loss in the cable $= 12.5 - 1.5 = 11.0$ mW $= 11.0 \times 10^{-3}$ W

Maximum resistance of one wire $= \dfrac{P}{I^2} = \dfrac{(11.0 \times 10^{-3}) \div 2}{(2.5 \times 10^{-3})^2} = 880\ \Omega$

Hence, maximum distance $= \dfrac{R_{\text{wire}}}{R_{\text{per metre}}} = \dfrac{880}{0.27} = 3260$ m

6. **[RVHS/H2/Prelims 2010/P2/5]**

a. The electromotive force (e.m.f.) of a source is defined as the amount of electrical energy per unit charge that is converted from other forms of energy when charge passes through the source.

The potential difference (p.d.) between two points in a circuit is defined as the amount of electrical energy per unit charge that is converted to other forms of energy when charge passes from one point to the other.

b. **i.** **1.** $R = \rho \dfrac{L}{A} = (9.8 \times 10^{-7}) \times \dfrac{16}{0.20 \times 10^{-6}} = 78.4\ \Omega$

2. $P = \dfrac{V^2}{R} = \dfrac{(240)^2}{78.4} = 735$ W

ii.

	switch A	switch B	switch C
lowest	OFF	ON	OFF
highest	ON	OFF	ON

7. **[VJC/H2/Prelims 2010/P3/4]**

a. The resistance of a circuit component is defined as the ratio of the potential difference across it to the current in it.

b. For density d of the material to remain the same,

original area: $A=\dfrac{m}{dL}$

new area: $A'=\dfrac{m}{d(3L)}=\dfrac{A}{3}$

Resistance of the wire:

before stretching: $R=\rho\dfrac{L}{A}$

after stretching: $R'=\rho\dfrac{(3L)}{\frac{A}{3}}=9\rho\dfrac{L}{A}$

Hence, $R'=9R=9\times(6.0)=54\ \Omega$

c. i. If the safe power rating is 0.40 W, the maximum safe voltage for:

the 1000-Ω resistor: $V_{1000\text{-}\Omega,\ \max}=\sqrt{PR_{1000\text{-}\Omega}}=\sqrt{(0.40)(1000)}=20\text{ V}$

the 160-Ω resistor: $V_{160\text{-}\Omega,\ \max}=\sqrt{PR_{160\text{-}\Omega}}\sqrt{(0.40)(160)}=8.0\text{ V}$

By potential divider principle:

If $V_{1000\text{-}\Omega}=20\text{ V}$:

$$20=\left(\frac{500}{500+160}\right)V_{XY}\Rightarrow V_{XY}=26.4\text{ V}$$

$$V_{160\text{-}\Omega}=\left(\frac{160}{500+160}\right)\times(26.4)=6.4\text{ V}<8.0\text{ V}$$

If $V_{160\text{-}\Omega}=8.0\text{ V}$:

$$8.0=\left(\frac{160}{500+160}\right)V_{XY}\Rightarrow V_{XY}=33\text{ V}$$

$$V_{1000\text{-}\Omega}=\left(\frac{500}{500+160}\right)\times(33)=25\text{ V}>20\text{ V}$$

Hence, the maximum safe voltage would be 26.4 V.

ii. If this potential difference were exceeded, one of the 1000-Ω resistors would most likely fail.

This is because when V_{XY} exceeds 26.4 V, the maximum safe power for the 1000-Ω would be exceeded first.

8. **[AJC/H2/Prelims 2010/P2/5]**

a. By potential divider rule, $V_{WX}=\dfrac{10}{10+40}E=\dfrac{E}{5}$

b. i. Potential at points X and Y are equal since there is no current flowing between X and Y.

ii. $V_{WX}=V_{WY}$

$$\frac{R_p}{R_p+20}E=\frac{E}{5}$$

$$5R_p=R_p+20$$

$$R_p=5.0\text{ k}\Omega$$

iii. 1. Direction: Y to X

2. Resistance of thermistor changes, resulting in a non-zero potential difference between X and Y.

9. **[CJC/H2/Prelims 2010/P3/6 (part)]**

a. Potential difference V across 4.0-Ω resistor = potential difference across AJ.

$$V=\left(\frac{R}{R+r}\right)E=\frac{4.0}{4.0+2.0}\times12=8.0\text{ V}$$

b. At balance length,

$$\frac{L_{AJ}}{L_{AB}}V_{AB}=V$$

$$\frac{64}{100}\times\frac{R_{AB}}{R_{AB}+2.0}\times20=8.0$$

$$R_{AB}=3.3\ \Omega$$

10. **[Dunman High/H2/Prelims 2010/P3/4]**

a. i. When S is closed, the current through R is $I=\dfrac{E}{R+r}$. Hence, the potential difference across R is:

$$V_R=IR=\frac{ER}{R+r}$$

ii. When S is open, $E=kL_0$ where k is a constant we need not be concerned with.

When S is closed, $V_R=kL$.

Substitute into the equation in **a.i.**,

$$kL=\frac{kL_0R}{R+r}$$

$$R+r=\frac{L_0}{L}R$$

$$r=\left(\frac{L_0}{L}-1\right)R$$

iii. $r=\left(\dfrac{82.3}{78.9}-1\right)\times12=0.52\ \Omega$

b. $R = 4.0\ \Omega$

From a.ii., $\frac{r}{R}=\frac{L_0-L}{L}$. Using a smaller value of R gives a larger value of $\frac{L_0-L}{L}$.

This will result in a larger change in length measurement (i.e. in the difference L_0-L) and hence a more reliable determination of r.

c. $\frac{L_0}{L}=\frac{r}{R}+1 \Rightarrow \frac{1}{L}=\frac{r}{L_0}\left(\frac{1}{R}\right)+\frac{1}{L_0}$

By plotting $\frac{1}{L}$ against $\frac{1}{R}$, a straight line graph can be obtained.

The value of r can be found by dividing the gradient $\frac{r}{L_0}$ by the vertical intercept $\frac{1}{L_0}$.

11. [MI/H2/Prelims 2010/P3/3]

a. i. See definitions in lecture notes.

ii. 1. $V=E-Ir$, hence the vertical intercept of the graph of V against I gives the value of E.
From the graph, vertical intercept = E = 6.0 V.

2. $P=IV=1.2\times4.2=5.0\text{ W}$

3. $r=-\text{gradient}=-\frac{6.0-0.0}{0.00-4.00}=-\left(\frac{6.0}{-4.00}\right)=1.5\ \Omega$

b. Since the total resistance in each of the two branches is the same, the current I in both branches is the same. However, the potential drop from a to c [$I\times(1.0\ \Omega)$] is less than that from a to d [$I\times(2.0\ \Omega)$].

Hence, potential at c is higher than that at d. Current will flow from c to d.

12. [NJC/H2/Prelims 2010/P3/4(part)]

b. i. The wire of uniform cross-section and carries a constant current supplied by the battery E. To measure the unknown E_x, the contact c is moved until the galvanometer reads zero. The potential difference between ac is proportional to the length ac and is equal to E_x.
To calibrate a potentiometer, one switches from E_x to a standard cell E_s and moves the slide to a point d on wire ab to obtain another zero galvanometer reading. Then E_x can be easily computed from the known E_s of the standard cell and the measured lengths ac and ad from E_x / E_s = ac / ad.

ii. The potentiometer can measure the terminal potential difference to high accuracy without drawing any current from the unknown source, thus eliminating the need to take into account the unknown internal resistance of the unknown source.

c. i. At the balance length ac,

$$E_X=\frac{\text{ac}}{\text{ab}}\times\frac{R_{ab}}{R_{ab}+\frac{1}{2}R_{rheostat}}\times E$$

$$2.0=\frac{\text{ac}}{100}\times\frac{600}{600+200}\times5.0$$

ac = 53 cm

ii. Right to left (across the cell E_x)

iii. At the new balance length ac',

$$E_X=\frac{\text{ac}'}{\text{ab}}\times\frac{R_{ab}}{R_{ab}+\left(\frac{40}{50}\times R_{rheostat}\right)}\times E$$

$$2.0=\frac{\text{ac}'}{100}\times\frac{600}{600+320}\times5.0$$

ac' = 61 cm

13. [NYJC/H2/Prelims 2010/P3/3]

a. $V=V_{5.0\Omega}=\frac{5.0}{5.0+300+0.5}\times10=0.16\text{ V}$

b. i. Total resistance in the circuit: $R=7.0+300+0.5=307.5\ \Omega$

Current in the main circuit: $I=\frac{E}{R}=\frac{10}{307.5}=0.033\text{ A}$

ii. At balance length,

$$\frac{\text{PR}}{1.2}\times\frac{7.0}{307.5}\times10=0.16367$$

PR = 0.86 m

iii. When the value of e.m.f. is of the order of millivolts, the fractional uncertainty of the answer will be large as the answer is small, therefore it is not precise.

14. [PJC/H2/Prelims 2010/P3/3]

a. i. When the slide contact S is moved all the way up to the top of the 20 kΩ resistance, V_{xs} = 0 V.
When the slide contact S is moved all the way down, V_{xs} can be found using the potential divider principle:

$$V_{xs}=\left(\frac{20}{20+4}\right)\times12=10\text{ V}$$

Hence, the voltage range of V_{XS} is 0 V to 10 V.

ii. When the slide contact S is moved all the way up to the top of the 20 kΩ resistance,
V_{sz} = 12 V − 0 = 12 V
When the slide contact S is moved all the way down,
V_{sz} = 12 V − 10 V = 2 V
Hence the voltage range of V_{SZ} is 2 V to 12 V.

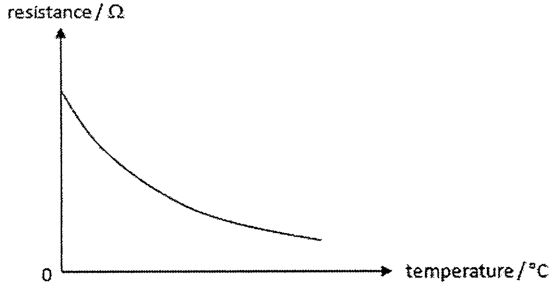
b. When the slide contact is placed at midpoint of the 20 kΩ track, R = 10 kΩ.
Effective resistance across XY when the voltmeter is connected:

$$R_{SY}=\frac{10\times10}{10+10}=5.0\text{ k}\Omega$$

Using potential divider principle, we have:

$$V_{SY}=\frac{5.0}{10+5.0+4.0}\times12\approx3.1579\text{ V}\approx3.2\text{ V (2 s.f.)}$$

The voltmeter reading = V_{SY} = 3.2 V c.



15. [YJC/H2/Prelims 2010/P2/2]

a. Let E be the e.m.f. of the test cell, and R the resistance of the resistance wire.

When the resistance of rheostat is 2.3 Ω, $E=\dfrac{0.57}{1.2}\times\dfrac{R}{R+1.5+2.3}\times 12$ ----(1)

When the resistance of rheostat is 3.5 Ω, $E=\dfrac{0.68}{1.2}\times\dfrac{R}{R+1.5+3.5}\times 12$ ----(2)

Solving (1) and (2), R = 2.4 Ω and E = 2.2 V.

b. There will be no change to the balance lengths.

c. If the resistance of rheostat is too high, the p.d. across the resistance wire will be too low. If the p.d. across the resistance wire is lower than the e.m.f. of the test cell, there will be no balance point.

16. [RI/H2/Prelims 2010/P3/2]

a. The resistance of a conductor is defined as the ratio of the potential difference across it to the current in it.

The resistivity of a conductor is the constant of proportionality relating its electrical resistance to the dimensions of the conductor (length and cross-sectional area).

b. $R=\rho\dfrac{L}{A}=\rho\dfrac{L}{\pi\times\dfrac{d^2}{4}}=\dfrac{4\rho}{\pi}\dfrac{L}{d^2}$

$$\frac{R_{AB}}{R_{BC}}=\frac{L_{AB}}{d_{AB}^2}\times\frac{d_{BC}^2}{L_{BC}}=\frac{50.0}{d^2}\times\frac{(0.30d)^2}{30.0}=0.15$$

c. i. Potential difference across the resistor r in the external circuit:

$$V_r=\frac{r}{2r}\times(2.00\text{ V})=1.00\text{ V}=V_{BJ}\text{ at null deflection}$$

Since balance length $L_{BJ}=\dfrac{1}{2}L_{BC}$, $V_{BC}=2V_{BJ}=2V_r=2.00$ V

ii. Method 1:

From **c.i.**, $V_{BC}=2.00$ V .

From **b.**, $\dfrac{R_{AB}}{R_{BC}}=\dfrac{V_{AB}}{V_{BC}}=0.15$. Hence,

$V_{AB}=0.15\times 2.00=0.30$ V

$V_{AC}=0.30+2.00=2.30$ V

OR $V_{AC}=1.15\times 2.00=2.30$ V

$E=V_{AC}+IR$

$2.50=2.30+0.400R$

$R=0.500\ \Omega$

Method 2:

From **c.i.**, $R_{BC}=\dfrac{V_{BC}}{I}=\dfrac{2.00}{0.400}=5.00\ \Omega$

From **b.**, $\dfrac{R_{AB}}{R_{BC}}=0.15$. Hence,

$R_{AB}=0.15\times 5.00=0.75\ \Omega$

$R_{AC}=0.75+5.00=5.75\ \Omega$

OR $R_{AC}=1.15\times 5.00=5.75\ \Omega$

$R_T=\dfrac{2.50}{0.400}=R_{AC}+R$

$R=0.500\ \Omega$

d. Over-estimate. 0.20 V is actually the p.d. across R as well as that of the ammeter. So,

$E-V_{AC}=I(R+R_A)$

$R+R_A=0.500\ \Omega$

$R<0.500\ \Omega$

17. [SAJC/H2/Prelims 2010/P3/4]

a. i. Since the two lamps are identical, the currents passing through them must be equal. Thus, current through **A₁** is $I=2\times 0.50=1.0$ A

Potential difference across **L₁**: $V_{L_1}=E-I_1r=2.0-(1.0\times 0.20)=1.8$ V

ii. Connecting L_3 parallel to L_1 and L_2 will lower the effective resistance of the circuit. Hence, current in A_1 increases.

b. i. By potential divider rule, $V_{AB}=\dfrac{2.0}{8.0+2.0}\times 2.00=0.40$ V

ii. $E_Y=V_{JB}=\dfrac{70.0}{100.0}\times 0.40=0.28$ V

iii. Replace the resistance wire with one of lower resistivity.

Replace Cell **X** with one of lower e.m.f.

Replace the 8.0 Ω resistor with one of higher resistance.